

BelkaCTF #7 — Compromised Air Traffic Control Workstation

Case INC-2026-0624-0001 · Examiner: niel · 2026-06-24 21:47 UTC

4 FINDINGS 1 EVIDENCE ITEMS 9 IOCS



CLASSIFICATION CONFIDENTIAL — AVIATION SAFETY IMPLICATIONS	TARGET SYSTEM TSO-ATC-CT412 (Windows 10 build 19041 (19041.1.amd64fre.vb_releas)	COMPROMISE DETECTED YES — 3 CRITICAL severity findings, active C2 established	INVESTIGATION WINDOW 2025-07-22 18:50 → 19:18 UTC (28 minutes)
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On 2025-07-22, an air traffic control workstation (TSO-ATC-CT412, user 'award') was compromised within 3 minutes of user login. A phishing email from ktso.sec@mailforce.net delivered a trojan (epxlorer.exe) masquerading as the Windows shell process explorer.exe. The malware established command-and-control to an attacker-operated Telegram Bot API behind Cloudflare infrastructure within 19 minutes of execution. At the time of compromise, ATC_RADAR.exe — an aviation radar processing application — was actively running, meaning unauthorized code coexisted with live air traffic operations. This constitutes a potential aviation safety incident with possible regulatory reporting obligations under ICAO Annex 17. Immediate network isolation of the affected workstation and escalation to the aviation safety officer are required.

Chronological reconstruction of events from memory forensics artifacts. Color-coded by event type. Hover cards for source details.

2025-07-22 18:50:46 UTC

System boot

volatility3 windows.info

2025-07-22 18:50:50 UTC

User 'award' logs in — explorer.exe (PID 5116) starts

volatility3 windows.pslist

2025-07-22 18:51:58 UTC

Microsoft Edge establishes HTTPS connection (browsing activity begins)

volatility3 windows.netscan

2025-07-22 18:53:11 UTC

⚠ MALWARE EXECUTION: explorer.exe (PID 9920) spawned by explorer.exe

F-niel-001

volatility3 windows.pslist + MemProcFS

2025-07-22 19:12:33 UTC

C2 ESTABLISHED: explorer.exe → 104.211.247:443 (Telegram Bot API)

F-niel-002

volatility3 windows.netscan

2025-07-22 19:15:05 UTC

ATC_RADAR.exe active — malware coexisting with aviation operations software

F-niel-004

volatility3 windows.pslist + MemProcFS

2025-07-22 19:18:53 UTC

Memory dump captured — evidence acquisition point

volatility3 windows.info

● Malware / Compromise ● Network / C2 ● Process Activity ● File System ● System Event ● User Activity



ID	FINDING	SEVERITY / CONFIDENCE	MITRE ATT&CK	TOOL	EVIDENCE
F-niel-001	Process epexplorer.exe (PID 9920) was spawned by explorer.exe (PID 5116) at 2025-07-22 18:53:11 UTC, just 2 minutes 21 seconds after user login. The binary was executed from C:\Users\award\Downloads\epexplorer.exe.	CRITICAL HIGH conf	T1036.005 T1204.002	volatility3 2.7.0 + MemProcFS 5.17.8	EVID-001
F-niel-002	epexplorer.exe established an outbound HTTPS connection to 104.211.247:443 at 2025-07-22 19:12:33 UTC — 19 minutes after execution. The destination IP is a Cloudflare edge node fronting the attacker-controlled domain api-telegram-org.ctf.do. The malware uses Telegram Bot API (token: 8169267144:AAFwkhmXh71SMVcvFLantjTIwIRbT9HdJdU) as its C2 channel, abusing legitimate cloud infrastructure to blend with normal HTTPS traffic.	CRITICAL HIGH conf	T1102.002 T1573	volatility3 2.7.0	EVID-001
F-niel-003	epexplorer.exe was discovered at three virtual addresses in the memory filescan, including a partial browser download artifact (epexplorer.exe435568.crdown	HIGH HIGH conf	T1566.001 T1204.002	volatility3 2.7.0	EVID-001

ID	FINDING	SEVERITY / CONFIDENCE	MITRE ATT&CK	TOOL	EVIDENCE
	load). The .crdownload extension is specific to Chromium-based browsers (Microsoft Edge in this environment — confirmed by extensive Edge artifacts throughout the filescan). Per CTF intelligence, the phishing email originated from ktso.sec@mailforce.net.				
F-niel-004	The compromised host (TSO-ATC-CT412, user 'award') runs ATC_RADAR.exe — an air traffic control radar application with multiple active processes and network connections to 8.8.8.8 (Google DNS-over-HTTPS). The workstation name (ATC-CT412) and application context confirm this is not a standard enterprise endpoint — it is an aviation operations system. Multiple ATC_RADAR.exe instances ran concurrently with explorer.exe, meaning the malware coexisted with active ATC operations.	CRITICAL HIGH conf	T1003.001	volatility3 2.7.0 + MemProcFS 5.17.8	EVID-001



TYPE	INDICATOR	CONTEXT	FINDING
SHA-256	3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35	Evidence archive hash	EVID-001
SHA-256	7b45e797acabaf592784bf7efceb979e3f036d5b9dd47f52280c17e38864c2dd	Memory dump hash	EVID-001
MD5	75ea94b54420c39dcd3d8ce574ba9d34	Malware hash (epxlorer.exe)	F-niel-001
Filename	epxlorer.exe	Malware binary	F-niel-001
File Path	C:\Users\award\Downloads\epxlorer.exe	Execution path	F-niel-001
IP	104.21.1.247	C2 server (Cloudflare)	F-niel-002
Domain	api-telegram-org.ctf.do	Attacker C2 domain	F-niel-002
Email	ktso.sec@mailforce.net	Phishing sender	F-niel-003
IP	172.17.4.12	Compromised host	F-niel-004



EVID-001 — BelkaCTF_7_CASE250722_KTSAERO.zip

SHA-256: 3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35

Source: https://dl.ctf.do/BelkaCTF_7_CASE250722_KTSAERO.zip

TOOL	VERSION	RUNTIME	VALIDATED
volatility3	2.7.0	docker	✓
memprocfs	5.17.8	host_fuse	✓
plaso	20240512	docker	✓
mft-tools	1.2.0.0	docker	✓
sift-vm	?	?	✓
session_canary	?	?	⚠️ DEGRADED



PROPERTY	VALUE
Hostname	TS0-ATC-CT412
Username	award
Os	Windows 10 build 19041 (19041.1.amd64fre.vb_release.1912)
Architecture	x64
Processors	4
Host Ip	172.17.4.12
Dump Time	2025-07-22 19:18:53 UTC
Kernel Base	0xf80617e00000
Nt Product Type	NtProductWinNt
Source Tool	volatility3 2.7.0 windows.info.Info

Source: volatility3 2.7.0 windows.info.Info

Immediate containment and eradication steps per finding. Prioritize CRITICAL severity items first.

 F-niel-001: Malware Execution: epxlorer.exe Masquerading as explorer.exe

Immediately isolate TSO-ATC-CT412 from all networks. Kill PID 9920. Quarantine C:\Users\award\Downloads\epxlorer.exe. Do NOT reboot before capturing full disk image — volatile evidence in memory will be lost.

 F-niel-002: Command and Control via Fraudulent Telegram Bot API

Block outbound HTTPS to api-telegram-org.ctf.do at network perimeter. Block Telegram Bot API endpoints if not business-justified. Deploy SSL inspection for traffic to unknown Cloudflare IPs. Check firewall logs for additional connections from 172.17.4.12 in the 18:50-19:18 UTC window.

 F-niel-003: Phishing Delivery Vector: Browser Download of Malicious Executable

Search mail server for emails from ktso.sec@mailforce.net — identify all recipients. Purge the phishing email from all inboxes. Block mailforce.net at email gateway. Initiate user awareness training for 'award' and ATC personnel. Review Edge browser security policies — consider disabling automatic execution of downloaded files.

 F-niel-004: Critical Asset Compromised: Air Traffic Control Workstation

Escalate to aviation safety officer and CISO immediately. This may require mandatory reporting to the national aviation authority. Ground the workstation — do not allow it to process live radar data until forensic imaging is complete and the system is rebuilt from trusted media. Engage the ATC software vendor (ATC_RADAR.exe) to verify software integrity. Review separation requirements between administrative workstations and operational ATC systems.



The following areas were not examined and represent uncertainty in the assessment. Each gap should be addressed before finalizing conclusions.

⚠ **Disk Image Not Analyzed**

No disk image was available. Without filesystem forensics, persistence mechanisms, additional dropped files, and full file-system timeline cannot be confirmed.

⚠ **Network Capture Unavailable**

No PCAP provided. Encrypted C2 traffic cannot be analyzed for data exfiltration. Lateral movement cannot be assessed.

⚠ **Registry Hives Not Examined**

User activity artifacts and malware persistence locations (Run keys, scheduled tasks, services) remain unexamined.

⚠ **Malware Not Reverse-Engineered**

Full capability assessment of epxlorer.exe (keylogging, screen capture, lateral movement, exfiltration) is pending.

⚠ **Supplementary Evidence Not Analyzed**

The 5.6 GB supplementary file (SHA-256: e42f5254...) containing Android artifacts was not analyzed.

⚠ **Single Host Analysis Only**

Compromise of additional systems on the network cannot be ruled out without broader endpoint telemetry.

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Hermes Forensics Agent · Report generated 2026-06-24 21:47 UTC

This report is auto-generated from structured case data. All findings require human review before finalization.